

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(3 points) Let $f(x) = \begin{cases} \sqrt{x+2} & x \leq 2 \\ x-3 & x > 2 \end{cases}$, find the limits that exist or explain why if it doesn't exist.

$$(a) \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \sqrt{x+2} = \sqrt{4} = 2$$

$$(b) \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x-3) = 2-3 = -1$$

$$(c) \lim_{x \rightarrow 2} f(x) \text{ doesn't exist since } \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

Problem 2.(3 points) Find

$$\lim_{x \rightarrow 4} \frac{x-4}{x^2-16}$$

$$= \lim_{x \rightarrow 4} \frac{(x-4)}{(x-4)(x+4)}$$

$$= \lim_{x \rightarrow 4} \frac{1}{x+4}$$

$$= \frac{1}{4+4} = \frac{1}{8}$$

Problem 3.(4 points) Find

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{-4x^2 + 5x + 6}{-x^2 + 3x + 7}}$$

$$= \sqrt{\lim_{x \rightarrow +\infty} \frac{-4x^2 + 5x + 6}{-x^2 + 3x + 7}}$$

$$= \sqrt{\lim_{x \rightarrow +\infty} \frac{\frac{-4x^2 + 5x + 6}{x^2}}{\frac{-x^2 + 3x + 7}{x^2}}} = \sqrt{\lim_{x \rightarrow +\infty} \frac{-4 + \frac{5}{x} + \frac{6}{x^2}}{-1 + \frac{3}{x} + \frac{7}{x^2}}}$$

$$= \sqrt{\frac{\lim_{x \rightarrow +\infty} (-4 + \frac{5}{x} + \frac{6}{x^2})}{\lim_{x \rightarrow +\infty} (-1 + \frac{3}{x} + \frac{7}{x^2})}} = \sqrt{\frac{-4}{-1}} = \sqrt{4} = 2$$